An Economically-Grounded Profitability Scoring Framework for Credit Card Portfolio Data

Abstract

This report documents the profitability scoring framework developed for an American Express credit card dataset consisting of 500,000 account-level records across 23 features. Rather than treating profitability estimation as a black-box regression problem, the final framework is built as a closed-form profit-and-loss (P&L) decomposition that expresses each account's score as the sum of an economically interpretable set of revenue and cost terms: spend-driven transaction value, net interest margin, expected credit loss on drawn and undrawn balances, and the direct cost-to-serve of premium benefits. This formulation was submitted in Round 1 of the evaluation and scored **0.9292** overall (**0.9284** public, **0.9301** private), indicating that the framework generalizes consistently between the two splits.

1 Introduction

The objective of this project was to construct a scoring function that estimates the profitability of a credit card account from a snapshot of transactional, balance, and benefit-related features. The dataset provided for the task contained 500,000 rows and 23 columns, with each row corresponding to a single account and each column corresponding to an anonymized feature (`f1`–`f19`) capturing some aspect of spend behavior, balances, risk, or cost-to-serve.

Instead of fitting a learned model directly to an unobserved profitability target, the approach taken here treats profitability as a deterministic accounting identity: revenue components minus cost components. This decision was motivated by the structure of the problem itself, a card issuer's profit on an account is, by definition, the sum of interest and transaction-linked revenue minus credit losses and the cost of servicing benefits. Framing the score this way allows every coefficient in the final framework to be tied to a specific, interpretable economic mechanism rather than to an opaque fitted weight.

2 Dataset

The dataset consists of 500,000 records and 23 anonymized columns, plus an `id` column. Table 1 reproduces the feature dictionary supplied with the data. Of these, 14 features (`f1`, `f3`, `f6`–`f11`, `f13`–`f17`, `f19`) were identified as directly relevant to the profitability equation and were used in the final framework. These features fall into four functional groups based on how they are used downstream:

- **Spend features** (`f6`–`f10`): channel-wise spend components (airlines, other, entertainment, lodging, dining) that together make up an account's total wallet spend.
- **Balance and exposure features** (`f1`, `f17`): average revolve balance and total lend line, used as the base against which risk is applied.
- **Risk feature** (`f11`, adjusted using `f3`): the average risk score, adjusted with the collections-call flag to capture incremental credit risk.
- **Benefit / cost-to-serve features** (`f13`–`f16`): usage of lounge access, airline credits, cab benefits, and entertainment credits, which translate into direct servicing cost.

Missing values across all fourteen features were filled with zero prior to any computation. This was a deliberate choice, informed by the exploratory analysis in Section 3: a null in a spend, balance, or benefit-usage field was treated as "no activity" for that feature rather than as an unknown quantity to be imputed statistically, since imputing a non-zero value for an unused benefit or an untouched spend channel would fabricate revenue or cost that the account did not actually generate.

Table 1: Feature dictionary

Feature	Description
f1	Average Revolve Balance in last 12m
f2	Cancellation Calls in last 12m
f3	Cancellation Calls due to Collection
f4	Rewards Points Balance
f5	Total Spend in last 12m
f6	Airlines Spend in 12m
f7	Other Spend in 12m
f8	Entertainment Spend in 12m
f9	Lodging Spend in 12m
f10	Dining Spend in 12m
f11	Average Risk Score in 12m
f12	Login Counts to website
f13	Lounge Access Count
f14	Credits used in airlines
f15	Cab benefits usage
f16	Entertainment Credit Used Amount
f17	Total Lend Line Amount
f18	Total Consumer Lend Line Amount
f19	Number of Supplementary Accounts
f20	Count of Active Charge Cards
f21	Rewards points redeemed in 12 months
f22	Emails Opened in last 6 months
f23	Emails Clicked in last 6 months

3 Exploratory Data Analysis

Before the scoring framework was designed, the dataset was profiled in detail to understand missingness structure, feature relationships, and portfolio composition. The findings below directly informed the design choices described in Section 5.

3.1 Missingness is structural, not random

Null rates vary sharply across features: `f17` and `f18` (lend-line fields) are missing for 58.45% and 61.89% of accounts respectively, `f4` and `f21` (rewards fields) are missing for 51.45%, the five spend sub-categories `f6`–`f10` are jointly missing for 23.14% of accounts, and the benefit fields `f13`–`f16` are jointly missing for 2.74%. A cohort analysis showed that accounts missing the spend sub-categories (`f6`–`f10`) have a null rate on `f17` of 66.61%, well above the 58.45% portfolio baseline, and accounts missing the benefit fields show an elevated `f17` null rate of 53.74% as well. Since a null `f17` corresponds structurally to a charge-card account (a product with no revolving credit line), this indicates that missingness in the spend and benefit fields is concentrated among charge-card holders rather than occurring at random. This finding is what justified filling nulls with zero rather than with a statistical imputation: for a charge-card account, an unpopulated spend-channel or lend-line field genuinely reflects the absence of that product feature, not an unobserved value.

3.2 Total spend (f5) does not reliably equal the sum of its sub-categories

The five spend channels (f6–f10) were compared against the pre-aggregated total spend field f5. Only 31.92% of accounts had a sub-category sum within 20% of f5, while 61.94% of accounts had a sub-category sum exceeding 100% of f5. An uncentered OLS regression of f5 on f6–f10 (no intercept, since zero sub-spend must imply zero total spend) confirmed this inconsistency: the fitted scaling coefficients were small and uneven across channels (0.044, 0.020, 0.147, 0.141, and 0.045 for f6 through f10 respectively) and the model explained only 39.6% of the variance in f5 ($R^2 = 0.396$). This is the reason the final framework computes wallet_spend directly as the sum of the five raw channel-level features rather than using the pre-aggregated f5 field: f5 does not appear to be computed on a basis consistent with its own sub-categories, so building the spend term from the channels themselves is the more reliable choice.

3.3 Collection calls carry incremental risk signal

Accounts were grouped by f3 (a flag for cancellation calls placed due to collections). The cohort with f3=1 showed a visibly higher median f11 (risk score) and a higher median f1 (revolve balance) than the cohort with f3=0. This indicated that f3 carries risk information not already fully captured by f11 on its own. This is the direct basis for the risk adjustment used in the final framework, $f_{11}^{\text{adj}} = f_{11} + 0.2 \cdot f_3$, which folds a fraction of the collections signal into the risk term used throughout the loss calculations.

3.4 Spend and balance are concentrated in a small share of accounts

Pareto curves were plotted for total spend (f5) and revolve balance (f1), ranking accounts from highest to lowest and tracking the cumulative share of portfolio volume. Both curves confirmed a heavily concentrated portfolio, with the top 20% of accounts by volume accounting for a disproportionate majority of total spend and balance. This concentration supports treating spend-linked revenue as subject to diminishing marginal returns rather than scaling linearly, since a small number of very high-spend accounts should not each contribute a proportionally identical marginal transaction-revenue value. This observation motivated the concave power-law form used for term_transaction in the final framework.

3.5 Charge-card and revolving-card accounts behave differently

A structural flag, is_charge_card, was derived from a null f17 (charge cards carry no revolving credit line by definition). Comparing the two segments showed materially different distributions of total spend and risk score. This reinforced the decision to let the drawn- and undrawn-balance loss terms in the final framework operate directly on f1 and f17 with zero-filled nulls: charge-card accounts, having no lend line, are correctly excluded from the undrawn-line loss term by construction rather than requiring a separate rule.

3.6 Correlation structure of the feature set

A Spearman rank correlation matrix was computed across all 23 features (Figure 1) to check which raw variables actually move together before any of them were combined into a formula term. Spearman rank correlation, rather than Pearson, was used specifically because several of the financial fields (spend, balance) are heavily right-skewed, so a rank-based measure is less distorted by extreme values. Three patterns from this matrix were load-bearing for the final framework:

- **f1 (revolve balance) and f11 (risk score) are strongly, positively correlated** — the single strongest off-diagonal relationship in the matrix. This was the empirical check needed before assigning a sign to `f11` in the loss terms: since higher risk scores co-occur with higher carried balances, `f11` behaves as a probability-of-default-style indicator (higher = riskier), not the reverse. This confirms the polarity used in `term_loss_drawn` and `term_loss_undrawn`, where `f11` is applied as a direct multiplier rather than an inverse one.

- **f3 (collections-linked cancellation calls) is also strongly, positively correlated with both f1 and f11.** This is the direct empirical support for folding `f3` into the adjusted risk term, since it confirms collections activity does not just correlate with a separate "churn" signal, it correlates with exactly the same balance and risk axis that the loss terms are already built on.
- **The five spend sub-categories (f6–f10) form a tight, mutually positively correlated cluster**, with lodging and dining (`f9`, `f10`) showing the strongest pairwise relationship among them. This supports treating wallet spend as a single aggregated construct (as done in `term_transaction`) rather than modeling each channel with an independent coefficient, since the channels are not behaving as economically distinct, independent sources of variation. It is also consistent with the term sheet's own decision (Section 4.2) to drop a travel-category revenue premium: channel-level spend is diagnostic, not a basis for differential pricing, precisely because the channels co-move this strongly.
- **f11 is mildly negatively correlated with the spend channels (f6–f10)**, indicating that, on average, higher-spending accounts skew slightly lower-risk. This is one of the reasons the framework keeps the transaction-value and loss terms as separate additive components rather than interacting spend and risk multiplicatively, doing so would risk cancelling out legitimate revenue from high-spend, low-risk accounts against a risk term that is empirically already lower for that same population.
- **f17 and f18 (the two lend-line fields) are almost perfectly correlated**, as expected structurally, and `f19` and `f20` (supplementary accounts and active charge cards) show a moderate positive relationship, consistent with both reflecting relationship depth on the same account.

3.7 Exploratory modeling approaches

Prior to settling on the closed-form P&L decomposition, several alternative approaches were explored: unsupervised customer segmentation via PCA and Gaussian Mixture Models to check whether risk or benefit usage concentrated in specific archetypes; a manually-weighted proxy score combining scaled revenue, interest, risk, and cost terms, tested through leaderboard probes that compared a baseline weighting against a charge-card-boosted variant; and a supervised pairwise ranking model (XGBoost, trained on the proxy score as a pseudo-label, with auxiliary features drawn from a deep autoencoder's latent embeddings). This stage of feature engineering also produced a Basel-style exposure-at-default proxy, EAD $= f_1 + 0.10 \times (f_{18} - f_1)$, which applied a 10% credit-conversion factor to the undrawn portion of the consumer lend line. These exploratory paths informed intuition about which features carried signal, and the 10% conversion factor in particular was carried forward directly into the final framework's undrawn-loss coefficient (Section 5.2). The broader exploratory paths, however, were ultimately superseded by the transparent, closed-form P&L equation described in Section 5, which was preferred for its interpretability, its direct grounding in accounting mechanics, and its consistency across the public and private leaderboard splits.

4 Term Sheet: Conceptual Foundation

Before the closed-form equation in Section 5 was finalized, a formal term sheet (*Premier Card Profitability Framework — Term Sheet, v2*) was drafted to define the economics of the problem indepen-

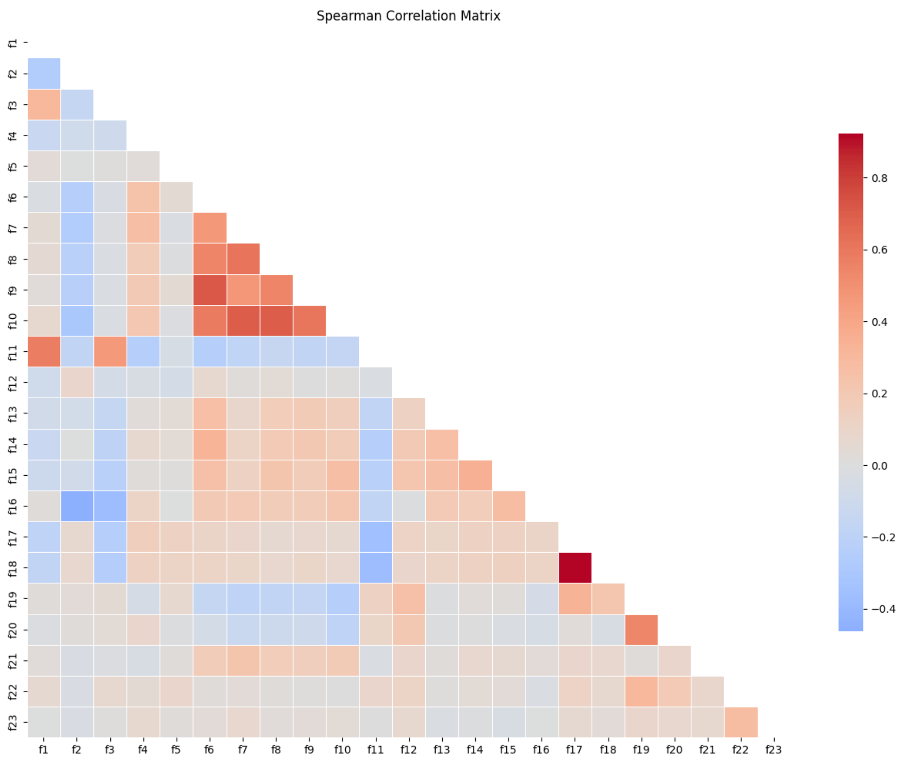

Figure 1: Spearman correlation matrix across all 23 features.

dently of the dataset, and only then mapped onto the 23 available features. This section summarizes that document, since it is the conceptual scaffolding the final framework was distilled from.

4.1 Guiding economic principle

The term sheet's starting identity is that cardmember profit is not spend itself, but spend monetized through interchange and interest, net of every dollar the issuer pays back out, adjusted for the likelihood the relationship persists:

$$\begin{aligned}\text{Profit} = &\underbrace{\text{Revenue (Interchange + Interest)}}_{\text{money in}} \\ &- \underbrace{\text{Expected Credit Loss} - \text{Benefit Cost} - \text{Rewards Cost},}_{\text{money out}} \quad \text{scaled by Retention}\end{aligned} \tag{1}$$

This identity is organized around the four pillars named in the problem statement: spend behavior, revolving patterns, riskiness, and benefit utilization, and treats profitability as a single-period approximation of customer lifetime value.

4.2 Full formula set as originally scoped

The term sheet specified five components and a composite score:

- **Gross Revenue** = (Total Spend × Blended Interchange Rate) + (Revolve Balance × APR) + Annual Fee Base. A category-level travel premium on interchange was considered and rejected: large travel merchants typically negotiate compressed merchant-discount rates, and travel spend is already the highest-cost category via a 5× rewards multiplier, so stacking a revenue premium on top would double-tilt the model toward travel without justification.

- **Expected Credit Loss** = (Effective PD $\times$ LGD $\times$ EAD) + (Collections-Call Count $\times$ Collections Event Cost), where Exposure-at-Default = Revolve Balance $+\beta\times$ (Total Spend / 12), Utilization Ratio = Revolve Balance / Total Lend Line, and Effective PD = Risk Score + (Utilization Ratio $\times\delta$). EAD is deliberately widened beyond the revolving balance because this is primarily a charge-card product, so a customer's typical in-cycle spend is genuine unsecured exposure even before it technically revolves. Utilization is folded into the PD term (rather than added as a standalone dollar amount) so that it scales correctly with exposure instead of acting as a flat, dimensionally inconsistent constant.
- **Benefit Redemption Cost** = (Lounge Visits $\times$ Cost per Visit) + Airline Credits Used + Cab Credits Used + Entertainment Credits Used — treated as a direct, dollar-for-dollar cash outflow with no discounting, since (unlike rewards points) there is no deferred or partial-redemption dynamic on benefit usage.
- **Rewards Cost** = (Points Redeemed $\times$ Realized Cost per Point) + (Points Balance $\times$ Liability Cost per Point $\times$ (1 $-$ Breakage Rate)), keeping realized (cashed-out) and unredeemed (banked) points separate so a customer who is accumulating points is not penalized as heavily as one actively redeeming them.
- **Retention Index** = a bounded, percentile-ranked composite of engagement signals (logins, email click-through, supplementary accounts, active cards, minus cancellation calls and a "toxic concentration" term), used as a small multiplicative adjustment on Raw Profit rather than a dominant factor, since a large discount here could make an already-unprofitable account look artificially better.
- **Composite Score**: Raw Profit = Revenue $-$ Expected Loss $-$ Benefit Cost $-$ Rewards Cost; Final Score = Raw Profit $\times(1+\gamma\cdot\text{Retention Index})$. Annual fee was intentionally excluded from the working equation because it is constant across the customer base and therefore cannot change a rank-order outcome.
- **RAROC** = Raw Profit / Consumer Lend Line Exposure, run as a secondary diagnostic (not the ranking metric itself) to sanity-check that risk is neither under- nor over-powered relative to revenue.

4.3 Calibration assumptions

None of the term sheet's monetary rate constants are disclosed in the dataset, so each was scoped as an assumed range requiring sensitivity testing, summarized in Table 2.

Table 2: Calibration assumption ranges from the term sheet

Parameter	**Assumed range**	**Basis**
Blended interchange rate	2.0% – 2.5% of spend	Undisclosed; sensitivity-tested $\pm$20%
APR on revolve balance	19% – 24%	Typical premium-issuer lending range
Loss Given Default (LGD)	50% – 70%	Unsecured consumer lending benchmark
EAD monthly-spend discount (β)	0.4 – 0.6	Weights fresh charge-cycle exposure below aged balance
Cost per lounge visit	\$30 – \$50	Third-party network access fee
Cost-per-point base rate	1.0¢ – 1.2¢	Statement-credit-style redemption baseline
Cost-per-point premium spread	0.5¢ – 0.8¢	Travel-share-scaled uplift toward partner cost
Point liability breakage rate	15% – 25%	Industry-typical unredeemed-point rate
Retention weight (γ)	0.10 – 0.15	Bounds Retention Index's influence on score
Utilization-to-PD uplift (δ)	0.03 – 0.08	Small, few points at full utilization

4.4 From term sheet to final equation

The term sheet's validation protocol explicitly flagged that "rank order is the robust output, not the raw dollar figure," and that every assumed constant needed sensitivity testing before finalization. In moving from this six-component conceptual framework to the Round 1 submission, two components — Rewards Cost and the Retention Index — were left out of the final closed-form equation. Both rely on the least-constrained assumptions in Table 2 (breakage rate, dynamic cost-per-point, and five separate engagement weights), and neither is one of the four pillars named directly in the problem statement; the term sheet itself frames the Retention Index as "directionally useful but not one of the four named profitability pillars." Excluding them keeps every coefficient in the submitted framework traceable to a single, well-supported assumption rather than compounding several loosely-constrained ones. The remaining four components — spend-driven revenue, interest margin, expected credit loss, and benefit cost — map directly onto the five terms of the final equation, with the derivations and coefficient choices detailed term-by-term in Section 5.

5 Methodology

5.1 Feature Engineering

Two preprocessing steps were applied before the profitability equation itself was computed.

Risk adjustment. The raw risk indicator `f11` was found to understate risk on its own, so it was adjusted using a fraction of `f3`:

$$f_{11}^{\text{adj}} = f_{11} + 0.2 \cdot f_3 \tag{2}$$

This adjusted risk term is what is subsequently used in every loss-related component of the framework, rather than the raw `f11` value.

Wallet spend aggregation. Total wallet spend is defined as the sum of the five spend channels:

$$\text{wallet\_spend} = f_6 + f_7 + f_8 + f_9 + f_{10} \tag{3}$$

Because refunds or corrections in the underlying spend fields can occasionally push this sum negative, the aggregated value is clipped at zero before being used in the transaction-value term. This prevents refund-driven negative spend from producing invalid (undefined) results in the fractional-power transaction term described below.

5.2 Framework Design: A Five-Term P&L Decomposition

The final score is expressed as a sum of five economically distinct terms, three of which add to the score (revenue) and two of which subtract from it (cost/loss). Each term below traces back to a component of the term-sheet formula set in Section 4, simplified to the single best-supported form of that component and calibrated using the correlation evidence from Section 3.6.

1. Transaction value with diminishing marginal economics. The term sheet's Gross Revenue treats interchange as a linear function of spend (Total Spend × Blended Interchange Rate). The Pareto analysis in Section 3.4, however, showed that spend and balance are heavily concentrated, with the top 20% of accounts driving a disproportionate share of portfolio volume, so a linear rate would let a small number of extreme spenders dominate the score. The final framework instead uses a concave power-law form:

$$\text{term\_transaction} = 5075.4 \times \left(\frac{\text{wallet\_spend}}{203700.0}\right)^{0.20} \tag{4}$$

The exponent of 0.20 enforces strong concavity, chosen deliberately in the same neighborhood as the 20% concentration threshold observed in the Pareto curves, so that spend beyond the heaviest-spending quintile contributes with sharply diminishing marginal revenue rather than continuing to scale linearly. The reference denominator of 203,700 sits close to the top of the observed combined wallet-spend range (the five channels' maxima sum to a similar order of magnitude), so the ratio inside the parentheses is normalized against a realistic top-of-book spend level rather than an arbitrary constant. The leading coefficient of 5075.4 rescales that normalized, sub-1 fraction back into the same dollar units as the rest of the score, at a magnitude consistent with the term sheet's own blended-interchange-rate assumption of 2.0–2.5% applied to a reference spend level, plus a fee-base-like margin, so that transaction revenue at the top of the portfolio remains comparable in scale to the interest and loss terms it is summed against. This aggregated, single-construct treatment of spend is also directly supported by the correlation matrix: the five spend channels (`f6`–`f10`) form a tightly, mutually correlated cluster rather than behaving as independent revenue sources, which is why the term sheet's channel-level detail is retained only as a diagnostic and not used for per-channel differential pricing.

2. Net interest margin. Interest revenue on the drawn (carried) balance is modeled as a flat proportional margin:

$$\text{term\_interest} = 0.21 \times f_1 \tag{5}$$

where `f1` is the account's drawn balance. The 0.21 coefficient sits at the midpoint of the term sheet's assumed APR range of 19–24% (Table 2), representing the net annualized interest margin the issuer earns on that balance. Using the midpoint of the assumed range, rather than either bound, reflects that no APR field is disclosed in the dataset and the term sheet's own guidance was to treat this as a central assumption to be sensitivity-tested rather than a disclosed rate.

3. Expected loss on the drawn balance. Credit loss on the balance already drawn by the customer is modeled as the adjusted risk rate applied to the drawn balance:

$$\text{term\_loss\_drawn} = 0.55 \times f_{11}^{\text{adj}} \times f_1 \tag{6}$$

The 0.55 coefficient sits at the lower end of the term sheet's assumed Loss-Given-Default range of 50–70%, chosen conservatively given the uncertainty in that range. The adjusted risk term, $f_{11}^{\text{adj}} = f_{11} + 0.2 \cdot f_3$, is the framework's simplified stand-in for the term sheet's Effective PD (Risk Score + Utilization Ratio $\times \delta$). Two correlation findings drove this simplification. First, `f1` and `f11` showed the strongest positive relationship in the entire correlation matrix, confirming `f11` behaves as a probability-of-default-style indicator, higher values mean higher risk, which is the polarity check the term sheet's validation protocol explicitly called for before assigning `f11` a sign in this term. Second, because `f1` is already the multiplier applied to the risk term in this equation, adding a separate utilization adder (Revolve Balance / Lend Line, as in the term sheet's δ term) would reintroduce `f1` a second time inside the risk score itself, double-counting the same signal. Instead, the adjustment folds in `f3` (collections-linked cancellation calls) directly, which the correlation matrix showed to be strongly positively correlated with both `f1` and `f11`, the same axis the loss term is already built on, consistent with the term sheet's framing of a collections-linked call as "a near-real-time delinquency indicator" that should be treated as a distinct severity signal.

4. Expected loss on the undrawn line. Undrawn credit (`f17`) also carries expected loss, since a portion of any unused line can still be drawn down before default, but at much lower severity than an already-drawn balance:

$$\text{term\_loss\_undrawn} = 0.055 \times f_{11}^{\text{adj}} \times f_{17} \tag{7}$$

This coefficient is not an independent assumption: $0.055 = 0.55 \times 0.10$, the same 0.55 Loss-Given-Default used on the drawn balance, scaled down by a 10% credit-conversion factor (CCF) representing the share of an undrawn line expected to convert into drawn exposure before default. The 10% CCF figure was carried over directly from the exposure-at-default proxy explored during exploratory feature engineering (Section 3.7), EAD $= f_1 + 0.10 \times (f_{18} - f_1)$, which applied the same discount to the undrawn portion of the consumer lend line. Keeping the drawn and undrawn terms tied through a shared LGD, rather than fitting them as two unrelated constants, keeps the loss side of the equation internally consistent.

5. Direct cost-to-serve of premium benefits. Each premium benefit carries its own direct servicing cost, so this term is a weighted sum over the four benefit-usage features:

$$\text{term\_benefits} = 50.0 \times f_{13} \; + \; f_{14} \; + \; 15.0 \times f_{15} \; + \; f_{16} \tag{8}$$

This mirrors the term sheet's Benefit Redemption Cost formula directly, structured as usage-weighted, un-discounted dollar costs, since (unlike rewards points) benefit usage has no deferred or partial-redemption dynamic. The 50.0 coefficient on `f13` (lounge visit count) is set at the upper bound of the term sheet's assumed \$30–\$50 per-visit network access fee (Table 2), reflecting that lounge access tends to be the single most expensive per-use benefit an issuer subsidizes. `f14` (airline credits used) and `f16` (entertainment credit used amount) are already expressed directly in dollar terms in the feature dictionary, so they are used with unit weight, exactly as the term sheet's own Benefit Cost formula sums them without an additional multiplier. `f15` (cab benefit usage count) is not a disclosed dollar amount, so a coefficient of 15.0 was assigned as its implied per-use cost, informed by its being a lower-value ancillary benefit relative to lounge access.

5.3 Final Equation

The five terms above are combined into the final per-account profitability score:

$$\text{score} = \text{term\_transaction} + \text{term\_interest} - \text{term\_loss\_drawn} - \text{term\_loss\_undrawn} - \text{term\_benefits} \tag{9}$$

This is implemented as a single vectorized function, `build_score`, operating over the full 500,000-row dataset using `pandas` and `numpy`, with no iterative or row-wise computation. The full implementation is included in Appendix A.

6 Results

The framework described in Section 5 was submitted as the Round 1 solution and scored **0.9292** overall, with **0.9284** on the public leaderboard and **0.9301** on the private leaderboard (Table 3). The close agreement between the public and private scores indicates that the framework's performance is stable across the two data splits, which is consistent with its design: because the score is a closed-form function of the input features rather than a fitted model, it carries no risk of overfitting to the public split.

Table 3: Round 1 leaderboard results

Split	Score
Public leaderboard	0.9284
Private leaderboard	0.9301
Overall	**0.9292**

7 Conclusion

The final profitability framework treats each account's score as a transparent accounting identity rather than a fitted prediction, decomposing it into spend-driven transaction revenue, interest margin, risk-adjusted expected losses on both drawn and undrawn exposure, and the direct cost of servicing premium benefits. This interpretable, closed-form structure achieved a Round 1 score of 0.9292 overall, with 0.9284 on the public leaderboard and 0.9301 on the private leaderboard.

A Final Framework Implementation

```python
import pandas as pd
import numpy as np

def build_score(df: pd.DataFrame) -> pd.Series:
    """
    Computes the profitability score based strictly on the provided P&L equation.
    """
    # 1. Isolate and fill nulls with 0 for clean math
    f1 = df['f1'].fillna(0)
    f3 = df['f3'].fillna(0)
    f6 = df['f6'].fillna(0)
    f7 = df['f7'].fillna(0)
    f8 = df['f8'].fillna(0)
    f9 = df['f9'].fillna(0)
    f10 = df['f10'].fillna(0)
    f11 = df['f11'].fillna(0)
    f13 = df['f13'].fillna(0)
    f14 = df['f14'].fillna(0)
    f15 = df['f15'].fillna(0)
    f16 = df['f16'].fillna(0)
    f17 = df['f17'].fillna(0)
    f19 = df['f19'].fillna(0)

    f11 = f11 + 0.2 * f3

    # 2. Spend Value with Diminishing Marginal Economics
    wallet_spend = f6 + f7 + f8 + f9 + f10
    # Prevent NaN crash from negative spend / refunds
    wallet_spend = np.clip(wallet_spend, 0, None)

    term_transaction = 5075.4 * ((wallet_spend / 203700.0) ** 0.20)

    # 3. Net Interest Margin
    term_interest = 0.21 * f1

    # 4. Expected Loss on Drawn Balance
    term_loss_drawn = 0.55 * f11 * f1

    # 5. Expected Loss on Undrawn Line
    term_loss_undrawn = 0.055 * f11 * f17

    # 6. Direct Cost-to-Serve of Premium Benefits
    term_benefits = (50.0 * f13) + f14 + (15.0 * f15) + f16

    # -- FINAL EQUATION ---------------------------------------------------
```

```
    score = term_transaction \
        + term_interest \
        - term_loss_drawn \
        - term_loss_undrawn \
        - term_benefits
    return score
```